

Unit-III

① Optimal solution is ~~5~~ 1 4 5 3 ~~2~~ 7 6

Job Seq.	Machine A		Machine B		Idle time	
	in	out	in	out	A	B
1	0	3	3	11	-	3
4	3	9	11	17	-	-
5	9	19	19	31	-	2
3	19	34	34	44	-	3
2	34	46	46	56	-	2
7	46	55	56	59	-	-
6	55	66	66	67	1	7

Total elapsed time = 67 hours

Idle time for machine A = 1 hour

Idle time for machine B = 17 hours

② $\min \{A_i\} = 6$, $\max \{C_i\} = 17$, $\max \{B_i\} = 6$
 Here $\min \{A_i\} \geq \max \{B_i\}$

Job	1	2	3	4	5	6
Machine G	11	11	11	11	11	15
Machine H	11	7	12	4	6	7

Optimal sequence: 3 1 ~~2~~ 4 5 6

Job	machine A		machine B		machine C		Idle time		
	in	out	in	out	in	out	A	B	C
3	0	6	6	11	11	18	-	6	11
1	6	14	14	17	18	26	-	3	-
2	14	21	21	25	26	29	-	4	-
6	21	30	30	32	36	37	-	5	7
5	30	40	40	41	41	46	-	4	4
4	40	49	49	51	51	53	4	8	5
								+2	

Total elapsed time = 53 hours

Idle time for machine A = 1 hours

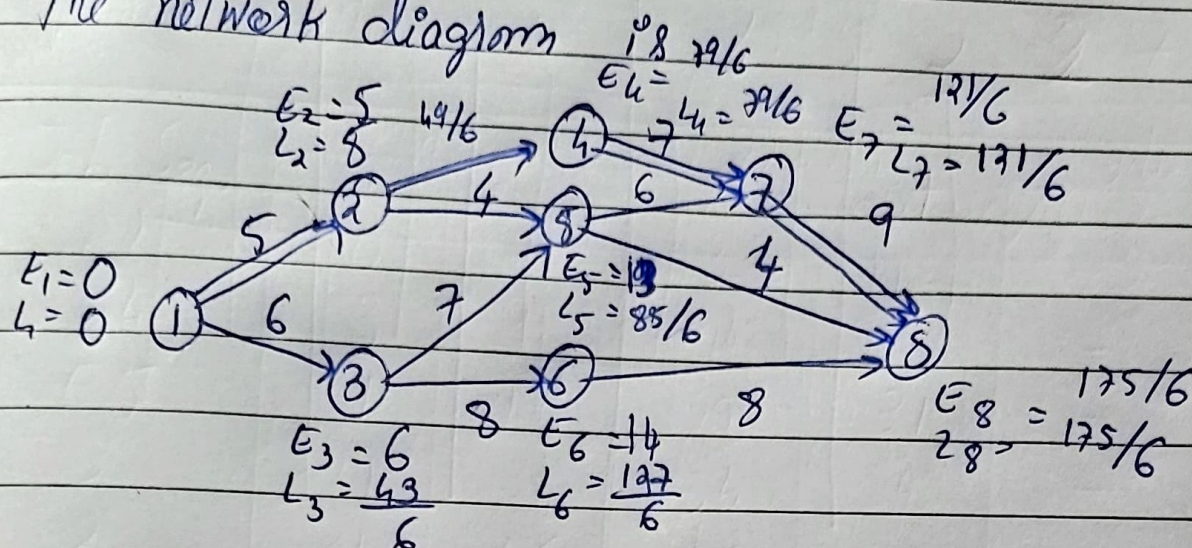
" " " " B = 32 hours

" " " " C = 27 hours

③ Activity	t_o	t_m	t_p	t_e	σ_e^2
1-2	3	5	7	5	$4/9$
1-3	2	6	10	6	-
3-5	6	7	8	7	-
2-5	3	4	5	4	-
2-4	7	8	10	$49/6$	$1/4$
3-6	6	8	10	8	-
4-7	4	7	10	7	1
5-7	5	6	7	6	-
5-8	3	4	5	4	-
6-8	7	8	9	8	-
7-8	7	9	11	9	$1/9$

Ans $t_e = \frac{t_o + 4t_m + t_p}{6}$, $\sigma_e^2 = \left(\frac{t_p - t_o}{6}\right)^2$

The network diagram



Critical Path: 1-2-4-7-8

Total project completion time = $5 + \frac{49}{6} + 7 + 9 = \frac{175}{6}$ days

$T_e = 29.1667$ day

Expected SD $\sigma_e = \sqrt{\frac{4}{9} + \frac{1}{9} + 1 + \frac{1}{9}} = 1.34$

$P(X \leq 30.1667) = P\left(Z \leq \frac{T_s - T_e}{\sigma_e}\right) = P\left(Z \leq \frac{30.1667 - 29.1667}{1.34}\right)$

$P(Z \leq 0.75) = 0.5 + \phi(0.75) = 0.5 + 0.2734 = 0.7734$